Name

Mathematics Extension 1 Year 11

The Binomial Theorem

Book 1 The Binomial Theorem

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Learning Intention ..... **Error! Bookmark not defined.**

Syllabus Content

ME1-11-05 uses the binomial theorem to solve problems and prove identities

- Recognise that a binomial is an expression with two terms, and that a binomial expansion is an expansion of a power of a binomial
- Examine the symmetry formed by the coefficients of decreasing powers of x in the expansion of $(x+y)^n$ for $n = 0,1,2,3,4,5$ and arrange the coefficients into Pascal's triangle
- Recognise the equivalence between the coefficient of $x^{n-r}y^r$ in the expansion of $(x+y)^n$ and ${}^nC_r$, when n is a positive integer
- Use patterns and symmetry in Pascal's triangle to confirm the identities ${}^nC_r = {}^{n-1}C_{r-1} + {}^{n-1}C_r$ for $1 \leq r \leq n-1$ and ${}^nC_r = {}^nC_{n-r}$ for $0 \leq r \leq n$
- Derive the binomial theorem: $(x+y)^n = {}^nC_0x^n + {}^nC_1x^{n-1}y + {}^nC_2x^{n-2}y^2 + \cdots + {}^nC_{n-1}xy^{n-1} + {}^nC_ny^n$, when n is a positive integer
- Apply the binomial theorem to expand and simplify expressions of the form $(x+y)^n$
- Use the binomial theorem to determine the coefficient of a term with a specific power or the constant term in a binomial expansion
- Use the identities ${}^nC_0 = 1$, ${}^nC_n = 1$, ${}^nC_r = {}^{n-1}C_{r-1} + {}^{n-1}C_r$ for $1 \leq r \leq n-1$ and ${}^nC_r = {}^nC_{n-r}$ for $0 \leq r \leq n$ to simplify expressions involving the binomial coefficients
- Prove identities involving binomial coefficients in binomial expansions by substituting values, comparing coefficients or applying a combinatorial argument to a specified context
- Apply given or proven identities involving binomial coefficients to prove further identities, without the use of calculus

Binomial Expansion, Pascal's Triangle, Combinations

Investigation Binomial expansion terms.

A **binomial** is an expression with **two terms**. The simplest of which is $(x + y)$.

Expand these expressions:

$(x + y)^0 =$

$(x + y)^1 =$

$(x + y)^2 =$

$(x + y)^3 =$

$(x + y)^4 =$

$(x + y)^5 =$

For a power of 0, there is 1 term.

For a power of 1, there are …… terms.

For a power of 2, there are …… terms.

For a power of n, there are …… terms.

Key Ideas

1. For $(x + y)^n$, the fully expanded and simplified result has …… terms.

Investigation Binomial expansion powers.

Consider the expansion of $(x+y)^5 = x^5 + 5x^4y + 10x^3y^2 + 10x^2y^3 + 5xy^4 + y^5$

If we arrange the terms in this way, and ignore the coefficients,

The 0th term is: x^5

The 1st term is: x^4y

The 2nd term is: ……

The 3rd term is: ……

The 4th term is ……

The 5th term is: ……

The powers of the terms always add up to ……

For $(x+y)^n$,

0th term: x^n

1st term: $x^{n-1}y$

2nd term: $x^{n-2}y^2$

k^{th} term: ……

n^{th} term: ……

The powers of the terms always add up to ……

Key Ideas

1. For $(x+y)^n$, each subsequent term …… the power of x by 1 and …… the power of y by 1, while the sum of the powers remains constant at ……

Investigation Binomial expansion and Pascal's Triangle

Pascal's triangle is a triangular pattern of numbers where each number equals the sum of the two numbers directly above it. For Pascal's triangle, **the first row is numbered 0**.

Continue the pattern to row 5.

Row 0	1
Row 1	1 1
Row 2	1 2 1
Row 3	1 3 3 1
Row 4	
Row 5	

What do you notice about the coefficients of binomial expansions on the previous page and the numbers in each row?

.....

.....

The connection between binomial expansion and Pascal's triangle:

Here is the expansion of $(x+y)^4$:

$$\begin{aligned}(x+y)^4 &= (x+y)(x^3+3x^2y+3xy^2+y^3)\\ &= x^4+3x^3y+3x^2y^2+xy^3+x^3y+3x^2y^2+3xy^3+y^4\\ &= 1x^4+4x^3y+6x^2y^2+4xy^3+1y^4\end{aligned}$$

Consider the $6x^2y^2$ term, here's where it comes from:

$$\begin{aligned}(x+y)^4 &= (x+y)(x^3+\mathbf{3x^2y}+\mathbf{3xy^2}+y^3)\\ &= x^4+3x^3y+\mathbf{3x^2y^2}+xy^3+x^3y+\mathbf{3x^2y^2}+3xy^3+y^4\\ &= 1x^4+4x^3y+\mathbf{6x^2y^2}+4xy^3+1y^4\end{aligned}$$

This happens for every term.

Key Ideas

1. For $(x+y)^n$, the ..... are the numbers in the corresponding row of Pascal's triangle.

Investigation Binomial expansion and Combinations

Consider $(x+y)^4 = 1x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + 1y^4$

Using a calculator, evaluate ${}^4C_0,\ {}^4C_1,\ {}^4C_2,\ {}^4C_3,\ {}^4C_4$

What do you notice?

……

The connection between binomial expansion and Pascal's triangle:

Think of $(x+y)^4$ as multiplying 4 sets of brackets: $(x+y)(x+y)(x+y)(x+y)$

Each term in the expansion comes from choosing x's or y's to multiply from these brackets.

x^4 term: Choose 4 x's from the 4 brackets: …… ways

x^3y term: Choose 3 x's from the 4 brackets (this automatically gives you 1 y): …… ways

x^2y^2 term: Choose 2 x's from the 4 brackets: …… ways

xy^3 term: Choose 1 x from the 4 brackets: …… ways

y^4 term: Choose 0 x from the 4 brackets: …… ways

Key Ideas

1. For $(x+y)^n$, the …… of each term are the ……
   ${}^nC_0,\ {}^nC_1,\ {}^nC_2, \ldots {}^nC_n$

Proof Pascal's Triangle and Combinations Identities.

For each of these combinations identities, describe how it is evident from the triangle.

n is the row, r is how much to go across a row. Remember that we start counting from 0.

1. ${}^nC_n = {}^nC_0 = 1$

Row 0 1
Row 1 1 1
Row 2 1 2 1
Row 3 1 3 3 1
Row 4 1 4 6 4 1
Row 5 1 5 10 10 5 1

2. ${}^nC_1 = {}^nC_{n-1} = n$

Row 0 1
Row 1 1 1
Row 2 1 2 1
Row 3 1 3 3 1
Row 4 1 4 6 4 1
Row 5 1 5 10 10 5 1

3. ${}^nC_r = {}^nC_{n-r}$ (symmetry property)

Row 0 1
Row 1 1 1
Row 2 1 2 1
Row 3 1 3 3 1
Row 4 1 4 6 4 1
Row 5 1 5 10 10 5 1

4. ${}^nC_r = {}^{n-1}C_{r-1} + {}^{n-1}C_r$ (Pascal's identity)

Row 0 1
Row 1 1 1
Row 2 1 2 1
Row 3 1 3 3 1
Row 4 1 4 6 4 1
Row 5 1 5 10 10 5 1

5. ${}^nC_0 + {}^nC_1 + {}^nC_2 + \ldots + {}^nC_n = 2^n$

Row 0 1
Row 1 1 1
Row 2 1 2 1
Row 3 1 3 3 1
Row 4 1 4 6 4 1
Row 5 1 5 10 10 5 1

Key Ideas

For $(x+y)^n$,

Number of terms:

The expansion has …… terms. We label them as term 0, term 1, term 2, up to term n.

Powers pattern:

Each term decreases the power of x by 1 and increases the power of y by 1.
The powers always add to ……

Coefficients:

The coefficient of each term follows the pattern: ……, ……, ……, ……, ⋯ ……

These match the numbers in row n of Pascal's triangle.

Using Row 5 of Pascal's triangle, derive the expansion of $(x+y)^6$ without expanding.

1 5 10 10 5 1

$(x+y)^6 =$

Write the expression for $(x+y)^n$ using what you have learned this lesson.
This is called the **Binomial Theorem.**

$(x+y)^n =$

Expanding Expressions using Binomial Theorem

ⓘ Binomial Theorem

$$(x+y)^n = {}^nC_0x^n + {}^nC_1x^{n-1}y + {}^nC_2x^{n-2}y^2 + \cdots + {}^nC_{n-1}xy^{n-1} + {}^nC_ny^n$$

n	${}^nC_0$	${}^nC_1$	${}^nC_2$	${}^nC_3$	${}^nC_4$	${}^nC_5$	${}^nC_6$	${}^nC_7$	${}^nC_8$	${}^nC_9$	${}^nC_{10}$
0	1										
1	1	1									
2	1	2	1								
3	1	3	3	1							
4	1	4	6	4	1						
5	1	5	10	10	5	1					
6	1	6	15	20	15	6	1				
7	1	7	21	35	35	21	7	1			
8	1	8	28	56	70	56	28	8	1		
9	1	9	36	84	126	126	84	36	9	1	
10	1	10	45	120	210	252	210	120	45	10	1

Example Expand expressions in the form $(x+y)^n$ using the binomial theorem.

Expand $(2+3x)^4$

$$(2+3x)^4 = {}^4C_0(2)^4 + {}^4C_1(2)^3(3x) + {}^4C_2(2)^2(3x)^2 + {}^4C_3(2)(3x)^3 + {}^4C_4(3x)^4$$
$$= 16 + 96x + 216x^2 + 216x^3 + 81x^4$$

Practice

Expand $(1+2a)^5$

$1 + 10a + 40a^2 + 80a^3 + 80a^4 + 32a^5$

Expand $(1-x)^6$

$1 - 6x + 15x^2 - 20x^3 + 15x^4 - 6x^5 + x^6$

Expand $\left(5x + \frac{a}{2}\right)^3$

$125x^3 + \frac{75x^2a}{2} + \frac{15xa^2}{4} + \frac{a^3}{8}$

Expand $\left(2x - \frac{3}{x}\right)^4$

Hint: write as $3x^{-1}$

$16x^4 - 96x^2 + 216 - \frac{216}{x^2} + \frac{81}{x^4}$

1 Use Pascal's triangle to expand:

a $(1 + x)^6$

$1 + 6x + 15x^2 + 20x^3 + 15x^4 + 6x^5 + x^6$

b $(1 - x)^6$

$1 - 6x + 15x^2 - 20x^3 + 15x^4 - 6x^5 + x^6$

c $(1 + x)^9$

$1 + 9x + 36x^2 + 84x^3 + 126x^4 + 126x^5 + 84x^6 + 36x^7 + 9x^8 + x^9$

d $(1 + 2y)^4$

$1 + 4(2y) + 6(4y^2) + 4(8y^3) + 16y^4$
$= 1 + 8y + 24y^2 + 32y^3 + 16y^4$

e $\left(1 - \frac{1}{x}\right)^8$

$= 1 - \frac{8}{x} + \frac{28}{x^2} - \frac{56}{x^3} + \frac{70}{x^4} - \frac{56}{x^5} + \frac{28}{x^6} - \frac{8}{x^7} + \frac{1}{x^8}$

f $(1 - 3z)^3$

$1 + 3(-3z) + 3(9z^2) + (-27z^3)$
$= 1 - 9z + 27z^2 - 27z^3$

g $\left(1 + \frac{2}{x}\right)^5$

$1 + 5\left(\frac{2}{x}\right) + 10\left(\frac{4}{x^2}\right) + 10\left(\frac{8}{x^3}\right) + 5\left(\frac{16}{x^4}\right) + \frac{32}{x^5}$
$= 1 + \frac{10}{x} + \frac{40}{x^2} + \frac{80}{x^3} + \frac{80}{x^4} + \frac{32}{x^5}$

h $\left(1 + \frac{y}{x}\right)^5$

$= 1 + \frac{5y}{x} + \frac{10y^2}{x^2} + \frac{10y^3}{x^3} + \frac{5y^4}{x^4} + \frac{y^5}{x^5}$

2 Use Pascal's triangle to expand:

a $(x + y)^4$

$x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4$

b $(2x + y)^5$

$32x^5 + 5(16x^4)y + 10(8x^3)y^2 + 10(4x^2)y^3 + 5(2x)y^4 + y^5$
$= 32x^5 + 80x^4y + 80x^3y^2 + 40x^2y^3 + 10xy^4 + y^5$

c $(3x + 2y)^4$

$81x^4 + 4(27x^3)(2y) + 6(9x^2)(4y^2) + 4(3x)(8y^3) + 16y^4$
$= 81x^4 + 216x^3y + 216x^2y^2 + 96xy^3 + 16y^4$

d $(r - s)^6$

$r^6 - 6r^5s + 15r^4s^2 - 20r^3s^3 + 15r^2s^4 - 6rs^5 + s^6$

e $\left(a - \frac{b}{2}\right)^3$

$a^3 + 3a^2\left(-\frac{b}{2}\right) + 3a\left(\frac{b^2}{4}\right) + \left(-\frac{b^3}{8}\right)$
$= a^3 - \left(\frac{3}{2}\right)a^2b + \left(\frac{3}{4}\right)ab^2 - \left(\frac{1}{8}\right)b^3$

f $\left(x + \frac{1}{x}\right)^6$

$x^6 + 6x^5\left(\frac{1}{x}\right) + 15x^4\left(\frac{1}{x^2}\right) + 20x^3\left(\frac{1}{x^3}\right) + 15x^2\left(\frac{1}{x^4}\right) + 6x\left(\frac{1}{x^5}\right) + \frac{1}{x^6}$
$= x^6 + 6x^4 + 15x^2 + 20 + \frac{15}{x^2} + \frac{6}{x^4} + \frac{1}{x^6}$

g $(a - b)^9$

$a^9 - 9a^8b + 36a^7b^2 - 84a^6b^3 + 126a^5b^4 - 126a^4b^5 + 84a^3b^6 - 36a^2b^7 + 9ab^8 - b^9$

h $\left(\frac{r}{2} + \frac{s}{3}\right)^5$

$\left(\frac{r^5}{32}\right) + 5\left(\frac{r^4}{16}\right)\left(\frac{s}{3}\right) + 10\left(\frac{r^3}{8}\right)\left(\frac{s^2}{9}\right) + 10\left(\frac{r^2}{4}\right)\left(\frac{s^3}{27}\right) + 5\left(\frac{r}{2}\right)\left(\frac{s^4}{81}\right) + \left(\frac{s^5}{243}\right)$
$= \frac{r^5}{32} + \frac{5r^4s}{48} + \frac{5r^3s^2}{36} + \frac{5r^2s^3}{54} + \frac{5rs^4}{162} + \frac{s^5}{243}$

3 Use Pascal's triangle to expand:

a $(1 + x^2)^4$

$1 + 4x^2 + 6x^4 + 4x^6 + x^8$

b $(1 - 3x^2)^3$

$1 - 9x^2 + 27x^4 - 27x^6$

c $(x^2 + 2y^3)^6$

$x^{12} + 6x^{10}(2y^3) + 15x^8(4y^6) + 20x^6(8y^9) + 15x^4(16y^{12}) + 6x^2(32y^{15}) + 64y^{18}$

$= x^{12} + 12x^{10}y^3 + 60x^8y^6 + 160x^6y^9 + 240x^4y^{12} + 192x^2y^{15} + 64y^{18}$

d $\left(x - \frac{1}{x}\right)^9$

$x^9 + 9x^8\left(-\frac{1}{x}\right) + 36x^7\left(\frac{1}{x^2}\right) + 84x^6\left(-\frac{1}{x^3}\right) + 126x^5\left(\frac{1}{x^4}\right) + 126x^4\left(-\frac{1}{x^5}\right) + 84x^3\left(\frac{1}{x^6}\right) + 36x^2\left(-\frac{1}{x^7}\right) + 9x\left(\frac{1}{x^8}\right) + \left(-\frac{1}{x^9}\right)$

$= x^9 - 9x^7 + 36x^5 - 84x^3 + 126x - \frac{126}{x} + \frac{84}{x^3} - \frac{36}{x^5} + \frac{9}{x^7} - \frac{1}{x^9}$

e $\left(\sqrt{x} + \sqrt{y}\right)^7$

$x^{\frac{7}{2}} + 7x^3y^{\frac{1}{2}} + 21x^{(\frac{5}{2})y} + 35x^2y^{\frac{3}{2}} + 35x^{(\frac{3}{2})y^2} + 21xy^{\frac{5}{2}} + 7x^{(\frac{1}{2})y^3} + y^{\frac{7}{2}}$

$= x^3\sqrt{x} + 7x^3\sqrt{y} + 21x^2y\sqrt{x} + 35x^2y\sqrt{y} + 35xy^2\sqrt{x} + 21xy^2\sqrt{y} + 7y^3\sqrt{x} + y^3\sqrt{y}$

f $\left(\frac{2}{x} + 3x^2\right)^5$

$\frac{32}{x^5} + 5\left(\frac{16}{x^4}\right)(3x^2) + 10\left(\frac{8}{x^3}\right)(9x^4) + 10\left(\frac{4}{x^2}\right)(27x^6) + 5\left(\frac{2}{x}\right)(81x^8) + 243x^{10}$

$= \frac{32}{x^5} + \frac{240}{x^2} + 720x + 1080x^4 + 810x^7 + 243x^{10}$

4 Without expanding, simplify:

a $1 + 3(x - 1) + 3(x - 1)^2 + (x - 1)^3$

$= (1 + (x - 1))^3 = x^3$

b $1 - 6(x + 1) + 15(x + 1)^2 - 20(x + 1)^3 + 15(x + 1)^4 - 6(x + 1)^5 + (x + 1)^6$

$= (1 - (x + 1))^6 = (-x)^6 = x^6$

5 Without expanding, simplify:

a $y^5 + 5y^4(x - y) + 10y^3(x - y)^2 + 10y^2(x - y)^3 + 5y(x - y)^4 + (x - y)^5$

$= (y + (x - y))^5 = x^5$

b $x^3 + 3x^2(2y - x) + 3x(2y - x)^2 + (2y - x)^3$

$= (x + (2y - x))^3 = (2y)^3 = 8y^3$

6 Find integers a and b such that:

a $(1 + \sqrt{3})^5 = a + b\sqrt{3}$

$1 + 5\sqrt{3} + 10(3) + 10(3\sqrt{3}) + 5(9) + 9\sqrt{3}$

$= 1 + 5\sqrt{3} + 30 + 30\sqrt{3} + 45 + 9\sqrt{3}$

$= 76 + 44\sqrt{3}$

a = 76, b = 44

b $(1 - \sqrt{5})^3 = a + b\sqrt{5}$

$1 + 3(-\sqrt{5}) + 3(5) + (-5\sqrt{5})$

$= 1 - 3\sqrt{5} + 15 - 5\sqrt{5}$

$= 16 - 8\sqrt{5}$

$a = 16, b = -8$

7 Verify by direct expansion, and by taking out the common factor, that:

$$(1 + x)^4 - (1 + x)^3 = x(1 + x)^3$$

Method 1: Taking out common factor

$(1 + x)^3((1 + x) - 1) = (1 + x)^3(x) = x(1 + x)^3$

Method 2: Direct expansion

$(1 + x)^4 = 1 + 4x + 6x^2 + 4x^3 + x^4$

$(1 + x)^3 = 1 + 3x + 3x^2 + x^3$

$(1 + x)^4 - (1 + x)^3 = x + 3x^2 + 3x^3 + x^4 = x(1 + 3x + 3x^2 + x^3) = x(1 + x)^3$

Finding a Particular Term

ⓘ General Term of a Binomial Expansion

The rth term in $(x+y)^n$ is ${}^nC_r x^{n-r} y^r$.

Identify r^{th} term in a binomial expansion.

Find the rth term:

a $(x+1)^7$	**b** $(3+2x)^5$
c $(2x-3)^4$	**d** $(x-4y)^6$
e $(x^2+x)^{11}$	**f** $\left(x^2+\frac{1}{x}\right)^4$
g $\left(x^3+\frac{2}{x}\right)^5$	**h** $\left(\frac{2}{x^3}-\frac{x}{3}\right)^8$

ⓘ General Term of a Binomial Expansion

The rth term in $(x+y)^n$ is ${}^nC_r x^{n-r} y^r$.

If you are asked to find a term with a particular power of x:

1. Find r^{th} term by substituting n and simplifying so there is only 1 x term.
2. Find the value of r by equating powers of x.
3. Evaluate r^{th} term.

Example Find a particular term of $(x+y)^n$ using the binomial theorem.

Find the term containing x^4 in $(2x-3)^6$

1. r^{th} term: ${}^6C_r(2x)^{6-r}(-3)^r$
   $= {}^6C_r(2^{6-r})(x^{6-r})(-3)^r$
2. For x^4, let $6-r=4$; $r=2$
3. $\therefore {}^6C_2(2^{6-2})(x^{6-2})(-3)^2 = 2160x^4$

Find the term independent of x in

$$\left(x^2+\frac{2}{x^2}\right)^8$$

1. r^{th} term: ${}^8C_r(x^2)^{6-r}(2x^{-2})^r$
   $= {}^8C_r x^{16-2r}2^r x^{-2r}$
   $= {}^8C_r x^{16-4r}2^r$
2. Let $16-4r=0$; $r=4$
3. $\therefore {}^8C_4 x^{16-4(4)}2^4 = 1120$

Practice

Find the term containing x^3 in $(3x-2)^5$

Find the term containing x^2 in $(1+3x^2)^5$

Find the term independent of x in $\left(2x+\frac{1}{x^2}\right)^6$

Find the term containing x^3y^4 in $(x+y)^7$

1 Write down the general term for each binomial expansion.

a $(1 + x)^{13}$

$^{13}C_k\, x^k$

b $(1 + 2x)^7$

$^{7}C_k\, 2^k\, x^k$

c $(5 + 7x)^{12}$

$^{12}C_k\, 5^{12-k}\, 7^k\, x^k$

d $(2x - y)^9$

$^{9}C_k\, 2^{9-k}\, (-1)^k\, x^{9-k}\, y^k$

e $(x + 2x^{-1})^5$

$^{5}C_k\, x^{5-k}\, 2^k\, x^{-k} = {}^{5}C_k\, 2^k\, x^{5-2k}$

f $\left(6x - \frac{2}{x}\right)^8$

$^{8}C_k\, (6x)^{8-k}\, (-2)^k\, x^{-k} = {}^{8}C_k\, 3^{8-k}\, (-1)^k\, 2^8\, x^{8-2k}$

2 For $(1 + x)^{11}$, the rth term is $^{11}C_r\, x^r$.

a find the term in x^2

For x^2, let $r = 2$
$\therefore\ {}^{11}C_2\, x^2 = 55x^2$

b find the term in x^8

For x^8, let $r = 8$
$\therefore\ {}^{11}C_8\, x^8 = 165x^8$

3 For $(1 - x)^7$

a find the term in x^3

$^{7}C_r\, (-1)^r x^r$
For x^3, let $r = 3$
$\therefore\ {}^{7}C_3\, (-1)^3\, x^3 = -35x^3$

b find the term in x^5

For x^5, let $r = 5$
$\therefore\ {}^{7}C_5\, (-1)^5\, x^5 = -21x^5$

4 For $(1 + 2x)^6$

a find the term in x^4

r^{th} term: $^{6}C_r\, 2^r x^r$
For x^4, let $r = 4$
$\therefore\ {}^{6}C_4\, 2^4\, x^4 = 240x^4$

b find the term in x^5

For x^5, let $r = 5$
$\therefore\ {}^{6}C_5\, 2^5\, x^5 = 192x^5$

5 For $\left(1 - \frac{3}{x}\right)^4$

a find the term in x^{-1}

r^{th} term: $^{4}C_r\, (-3)^r x^{-r}$
For x^{-1}, let $r = 1$
$\therefore\ {}^{4}C_1\, (-3)^1\, x^{-1} = -\frac{12}{x}$

b find the term in x^{-2}

For x^{-2}, let $r = 2$
$\therefore\ {}^{4}C_2\, (-3)^2\, x^{-2} = \frac{54}{x^2}$

6 In the expansion of $(1 + x)^6$:

a find the term in x^2,

$^6C_2\, x^2 = 15x^2$

b find the term in x^3,

$^6C_3\, x^3 = 20x^3$

c find the ratio of the term in x^2 to the term in x^3,

$\frac{15x^2}{20x^3} = \frac{3}{4x}$

7 In the expansion of $\left(1 + \frac{2}{3x}\right)^7$:

a find the term in x^{-5},

r^{th} term: $^7C_r \left(\frac{2}{3x}\right)^r = {}^7C_r\, 2^r\, 3^{-r}\, x^{-r}$

For x^{-5}, let $r = 5$

$^7C_5\, 2^5 3^{-5} x^{-5} = \frac{224}{81x^5}$

b find the term in x^{-6},

x^{-6}, let $r = 6$

$^7C_6\, 2^6 3^{-6} x^{-6} = \frac{448}{729x^6}$

c find the ratio of the term in x^{-5} to the term in x^{-6},

$\frac{9x}{2}$

8 Find the coefficient of x^4 in the expansion of $(1 - x)^4 + (1 - x)^5 + (1 - x)^6$.

Coefficient of x^4 in $(1 - x)^4$: $^4C_4\, (-1)^4 = 1$

Coefficient of x^4 in $(1 - x)^5$: $^5C_4\, (-1)^4 = 5$

Coefficient of x^4 in $(1 - x)^6$: $^6C_4\, (-1)^4 = 15$

Total: 1 + 5 + 15 = 21

9

a Expand $(1 + x)^4$ as far as the term in x^2.

$(1 + x)^4 = 1 + 4x + 6x^2 + \ldots$

b Hence find the coefficient of x^2 in the expansion of $(1 - 5x)(1 + x)^4$.

$(1 - 5x)(1 + x)^4 = (1 - 5x)(1 + 4x + 6x^2 + \ldots)$

Coefficient of x^2: $6 + (-5)(4) = -14$

10

a Expand $(1 + 2x)^5$ as far as the term in x^3.

$(1 + 2x)^5 = 1 + 10x + 40x^2 + 80x^3 + \ldots$

b Hence find the coefficient of x^3 in the expansion of $(2 - 3x)(1 + 2x)^5$.

$(2 - 3x)(1 + 2x)^5 = (2 - 3x)(1 + 10x + 40x^2 + 80x^3 + \ldots)$

Coefficient of x^3: $2(80) + (-3)(40) = 40$

11

a Expand $(1 - 3x)^4$ as far as the term in x^3.

$(1 - 3x)^4 = 1 - 12x + 54x^2 - 108x^3 + \ldots$

b Hence find the coefficient of x^3 in the expansion of $(2 + x)^2(1 - 3x)^4$.

$(4 + 4x + x^2)(1 - 12x + 54x^2 - 108x^3 + \ldots)$

Coefficient of x^3: $4(-108) + 4(54) + 1(-12) = -228$

12 Consider the expansion $\left(x^2 + \frac{1}{x}\right)^9$.

a Show that each term in the expansion of $\left(x^2 + \frac{1}{x}\right)^9$ can be written as ${}^9C_j\, x^{18-3j}$.

${}^9C_j\,(x^2)^{9-j}\,(x^{-1})^j$

$= {}^9C_j\, x^{18-2j}\, x^{-j}$

$= {}^9C_j\, x^{18-3j}$

b Hence find the coefficient of x^3

For x^3, let $18 - 3j = 3$, so $j = 5$

Coefficient: ${}^9C_5 = 126$

c Find the coefficient of x^{-3}

For x^{-3}, let $18 - 3j = -3$, so $j = 7$

Coefficient: ${}^9C_7 = 36$

d Find the term independent of x

For x^0, let $18 - 3j = 0$, so $j = 6$

Coefficient: ${}^{9}C_{6} = 84$

13

a Expand $(4 + x)^5$ as far as the term in x^3

$(4 + x)^5 = 1024 + 1280x + 640x^2 + 160x^3 + \ldots$

b Hence find the coefficient of x^3 in the expansion of $(3 - x)(4 + x)^5$.

$(3 - x)(4 + x)^5 = (3 - x)(1024 + 1280x + 640x^2 + 160x^3 + \ldots)$

Coefficient of x^3: $3(160) + (-1)(640) = -160$

14

a Expand $(1 - 2x)^6$ as far as the term in x^4.

$(1 - 2x)^6 = 1 - 12x + 60x^2 - 160x^3 + 240x^4 - \ldots$

b Hence find the coefficient of x^4 in the expansion of $(1 - 3x)(1 - 2x)^6$.

$(1 - 3x)(1 - 2x)^6 = (1 - 3x)(1 - 12x + 60x^2 - 160x^3 + 240x^4 - \ldots)$

Coefficient of x^4: $1(240) + (-3)(-160) = 720$

15 Find the specified terms in each expansion.

a For $(2 + x)^7$, find the term in x^2.

${}^{7}C_{r}\, 2^{7-r}\, x^{r}$

For x^2, let $r = 2$

$C_2\, 2^5\, x^2 = 672x^2$

b For $\left(x + \frac{y}{2}\right)^{14}$, find the term in x^9y^5.

${}^{14}C_{r}\, x^{14-r} \left(\frac{y}{2}\right)^{r}$

$= {}^{14}C_{r} \left(\frac{1}{2}\right)^{r} x^{14-r} y^{r}$

For x^9y^5, let $14 - r = 9$, so $r = 5$

${}^{14}C_{5} \left(\frac{1}{2}\right)^{5} x^9y^5 = \frac{1001x^9y^5}{16}$

c For $\left(\frac{x}{2} - 3y^2\right)^{11}$, find the term in $x^{10}y^2$.

${}^{11}C_{r} \left(\frac{x}{2}\right)^{11-r} (-3y^2)^{r}$

$= {}^{11}C_{r} \left(\frac{1}{2}\right)^{11-r} (-3)^{r} x^{11-r} y^{2r}$

For $x^{10}y^2$, let $11 - r = 10$ and $2r = 2$, so $r = 1$

${}^{11}C_{1} \left(\frac{1}{2}\right)^{10} (-3)^{1} x^{10}y^2$

$= -\frac{33x^{10}y^2}{1024}$

d For $\left(a - \sqrt{b}\right)^{20}$, find the term in a^2b^9.

${}^{20}C_{r}\, a^{20-r} \left(-b^{\frac{1}{2}}\right)^{r} = {}^{20}C_{r}\, (-1)^{r}\, a^{20-r}\, b^{\frac{r}{2}}$

For a^2b^9, let $20 - r = 2$ and $\frac{r}{2} = 9$, so $r = 18$

${}^{20}C_{18}\, a^2b^9 = 190a^2b^9$

16 Find the coefficient of:

a x^3 in $(3 - 4x)(1 + x)^4$

$(1 + x)^4 = 1 + 4x + 6x^2 + 4x^3 + \ldots$

$(3 - 4x)(1 + x)^4 = (3 - 4x)(1 + 4x + 6x^2 + 4x^3 + \ldots)$

Coefficient of x^3: $3(4) + (-4)(6) = -12$

b x in $(1 + 3x + x^2)(1 - x)^3$

$(1 - x)^3 = 1 - 3x + 3x^2 - x^3$

$(1 + 3x + x^2)(1 - x)^3 = (1 + 3x + x^2)(1 - 3x + 3x^2 - x^3)$

Coefficient of x: $1(-3) + 3(1) = 0$

c x^4 in $(5 - 2x^3)(1 + 2x)^5$

$(1 + 2y)^5 = 1 + 10y + 40y^2 + 80y^3 + 80y^4 + \ldots$

$(5 - 2x^3)(1 + 2y)^5 = (5 - 2x^3)(1 + 10y + 40y^2 + 80y^3 + 80y^4 + \ldots)$

Coefficient of x^4: $5(80) = 380$

d x^0 in $\left(1 - \frac{x}{3}\right)^3 \left(1 + \frac{2}{x}\right)^2$

$\left(1 - \frac{x}{3}\right)^3 = 1 - x + \left(\frac{x^2}{3}\right) - \left(\frac{x^3}{27}\right)$

$\left(1 + \frac{2}{x}\right)^2 = 1 + \left(\frac{4}{x}\right) + \left(\frac{4}{x^2}\right)$

$\left(1 - \frac{x}{3}\right)^3 \left(1 + \frac{2}{x}\right)^2 = \left(1 - x + \left(\frac{x^2}{3}\right) - \left(\frac{x^3}{27}\right)\right)\left(1 + \left(\frac{4}{x}\right) + \left(\frac{4}{x^2}\right)\right)$

Coefficient of x^0: $1(1) + (-1)\left(\frac{4}{x}\right)(x) + \left(\frac{1}{3}\right)\left(\frac{4}{x^2}\right)(x^2) = 1 - 4 + \frac{4}{3} = -\frac{5}{3}$

17 In the expansion of $(2x^3 + 3x^{-2})^{10}$, the general term is ${}^{10}C_k\,(2x^3)^{10-k}\,(3x^{-2})^k$.

a Show that this general term can be written as ${}^{10}C_k\,2^{10-k}\,3^k\,x^{30-5k}$.

$${}^{10}C_k\,(2x^3)^{10-k}\,(3x^{-2})^k$$
$$= {}^{10}C_k\,2^{10-k}\,x^{3(10-k)}\,3^k\,x^{-2k}$$
$$= {}^{10}C_k\,2^{10-k}\,3^k\,x^{30-3k-2k}$$
$$= {}^{10}C_k\,2^{10-k}\,3^k\,x^{30-5k}$$

b Hence find the coefficients of these terms, giving your answers in combination notation and as prime factors.

i x^{10}

For x^{10}, let $30 - 5k = 10$, so $k = 4$

${}^{10}C_4 \times 2^6 \times 3^4$

ii x^{-5}

For x^{-5}, let $30 - 5k = -5$, so $k = 7$

${}^{10}C_7 \times 2^3 \times 3^7$

iii x^0

For x^0, let $30 - 5k = 0$, so $k = 6$

${}^{10}C_6 \times 2^4 \times 3^6$

18

a Show that the general term in the expansion of $\left(\frac{x}{2} - \frac{5}{x}\right)^{15}$ is ${}^{15}C_j\,(-1)^j\,5^j\,2^{j-15}\,x^{15-2j}$.

$${}^{15}C_j\left(\frac{x}{2}\right)^{15-j}\left(-\frac{5}{x}\right)^j$$
$$= {}^{15}C_j\,x^{15-j}\,2^{-(15-j)}\,(-5)^j\,x^{-j}$$
$$= {}^{15}C_j\,(-1)^j\,5^j\,2^{j-15}\,x^{15-2j}$$

b Hence find, without simplifying, the coefficients of:

i x^{11}

For x^{11}, let $15 - 2j = 11$, so $j = 2$

${}^{15}C_2 \times 5^2 \times 2^{-13}$

ii x

For x, let $15 - 2j = 1$, so $j = 7$

$-{}^{15}C_7 \times 5^7 \times 2^{-8}$

iii x^{-5}

For x^{-5}, let $15 - 2j = -5$, so $j = 10$

${}^{15}C_{10} \times 5^{10} \times 2^{-5}$

19

a Find x if the terms in x^{10} and x^{11} in the expansion of $(5 + 2x)^{15}$ are equal.

Term in x^{10}: ${}^{15}C_{10}\, 5^5\, 2^{10}\, x^{10}$

Term in x^{11}: ${}^{15}C_{11}\, 5^4\, 2^{11}\, x^{11}$

Equating coefficients: ${}^{15}C_{10}\, 5^5\, 2^{10} = {}^{15}C_{11}\, 5^4\, 2^{11}$

${}^{15}C_{10}\, 5 = {}^{15}C_{11}\, 4$

Using ${}^{15}C_{10} = 3003$ and ${}^{15}C_{11} = 1365$

$3003(5) = 1365(4)x$

$x = \frac{11}{2}$

b Find x if the terms in x^{13} and x^{14} in the expansion of $(2 - 3x)^{17}$ are equal.

Term in x^{13}: ${}^{17}C_{13}\, 2^4\, (-3)^{13}\, x^{13}$

Term in x^{14}: ${}^{17}C_{14}\, 2^3\, (-3)^{14}\, x^{14}$

Equating coefficients: ${}^{17}C_{13}\, 2^4\, (-3)^{13} = {}^{17}C_{14}\, 2^3\, (-3)^{14}$

${}^{17}C_{13}\, 2(-1) = {}^{17}C_{14}\, 3$

Using ${}^{17}C_{13} = 2380$ and ${}^{17}C_{14} = 680$

$-2380(2) = 680(3)x$

$x = -\frac{7}{3}$

20 Determine the value of the term independent of x in the expansion of:

$$(1 + 2x)^4 \left(1 - \frac{1}{x^2}\right)^6$$

General term of $(1 + 2x)^4$: ${}^4C_r\, 2^r\, x^r$

General term of $\left(1 - \frac{1}{x^2}\right)^6$: ${}^6C_s\, (-1)^s\, x^{-2s}$

For x^0, need $r - 2s = 0$, so $r = 2s$

Possible: r = 0, s = 0: ${}^4C_0\, {}^6C_0 = 1$

r = 2, s = 1: ${}^4C_2\, 2^2\, {}^6C_1\, (-1)^1 = -144$

r = 4, s = 2: ${}^4C_4\, 2^4\, {}^6C_2\, 1 = 240$

Total: $1 - 144 + 240 = 97$

21 Find the coefficient of the power of x specified in each expansion.

a x^{15} in $\left(x^3 - \frac{2}{x}\right)^9$

$^9C_r\,(x^3)^{9-r}\left(-\frac{2}{x}\right)^r = {}^9C_r\,(-2)^r x^{27-3r-r} = {}^9C_r\,(-2)^r x^{27-4r}$

For x^{15}, let $27 - 4r = 15, so\ r = 3$

$^9C_3\,(-2)^3 = -672$

b x^{-14} in $(x - 3x^{-4})^{11}$

$^{11}C_r\,x^r\,(-3x^{-4})^{11-r} = {}^{11}C_r\,(-3)^{11-r}\,x^{r-44+4r} = {}^{11}C_r\,(-3)^{11-r}\,x^{5r-44}$

For x^{-14}, let $5r - 44 = -14, so\ r = 6$

$^{11}C_6\,(-3)^5 = -112\,266$

c the constant term in $\left(\frac{1}{2x^3} + x\right)^{20}$

$^{20}C_r\left(\frac{1}{2x^3}\right)^{20-r}x^r$

$= {}^{20}C_r\,2^{-(20-r)}x^{-60+3r+r}$

$= {}^{20}C_r\,2^{r-20}\,x^{4r-60}$

For x^0, let $4r - 60 = 0$, so $r = 15$

$^{20}C_{15}\,2^{-5} = \frac{969}{2}$

d x^7 in $\left(5x^2 + \frac{1}{2x}\right)^8$

$^8C_r\,(5x^2)^{8-r}\left(\frac{1}{2x}\right)^r = {}^8C_r\,5^{8-r}\,2^{-r}\,x^{16-2r-r} = {}^8C_r\,5^{8-r}\,2^{-r}\,x^{16-3r}$

For x^7, let $16 - 3r = 7, so\ r = 3$

$^8C_3\,5^5\,2^{-3} = 21\,875$

e x^{-1} in $\left(x^{-1} + \frac{2x}{7}\right)^5$

$^5C_r\,(x^{-1})^{5-r}\left(\frac{2x}{7}\right)^r = {}^5C_r\left(\frac{2}{7}\right)^r x^{-5+r+r} = {}^5C_r\left(\frac{2}{7}\right)^r x^{2r-5}$

For x^{-1}, let $2r - 5 = -1$, so $r = 2$

$^5C^2\left(\frac{2}{7}\right)^2 = \frac{40}{49}$

22 Find the coefficient of x in the expansion of $\left(x + \frac{1}{x}\right)^5\left(x - \frac{1}{x}\right)^4$.

This is $\left(x + \frac{1}{x}\right)\left(x^2 - \frac{1}{x^2}\right)^4$

$= (x + x^{-1})(x^2 - x^{-2})^4$

For $(x^2 - x^{-2})^4$, $^4C_r(x^2)^r(-x^{-2})^{4-r}$

$= {}^4C_r(-1)^{4-r}x^{4r-8}$

When multiplied by $(x + x^{-1})$, terms in x^0 and x^2 will become x^1

$4r - 8 = 0, r = 2$

$4r - 8 = 2, r = 2.5$ not possible

$\therefore {}^4C_2 = 6$

23 In the expansion of $(2 + ax + bx^2)(1 + x)^{13}$, the coefficients of x^0, x^1 and x^2 are all equal to 2. Find the values of a and b.

$(1 + x)^{13} = 1 + 13x + 78x^2 + \ldots$

$(2 + ax + bx^2)(1 + x)^{13}$:

Coefficient of x^0: $2(1) = 2$

Coefficient of x^1: $2(13) + a(1) = 2,$

$a = -24$

Coefficient of x^2: $2(78) + a(13) + b(1) = 2,$

$b = 158$

24 The expression $(1 + ax)^n$ is expanded in increasing powers of x.
Find the values of a and n if the first three terms are:

a $1 + 28x + 364x^2 + \ldots$

$(1 + ax)^n = 1 + nax + \left(\frac{n(n-1)}{2}\right)a^2x^2 + \ldots$

Coefficient of x: $na = 28$

Coefficient of x^2: $\left(\frac{n(n-1)}{2}\right)a^2 = 364$

Dividing both sides by na: $\frac{(n-1)a}{2} = 13,$

$(n-1)a = 26$

From $na = 28$: $na - a = 26$, so $a = 2, n = 14$

b $1 - \frac{10x}{3} + 5x^2 - \ldots$

$(1 + ax)^n = 1 + nax + \left(\frac{n(n-1)}{2}\right)a^2x^2 + \ldots$

Coefficient of x: $na = -10/3$

Coefficient of x^2: $\left(\frac{n(n-1)}{2}\right)a^2 = 5$

Dividing: $\frac{(n-1)a}{2} = \frac{5}{-\frac{10}{3}} = -\frac{3}{2}$, so $(n-1)a = -3$

From $na = -\frac{10}{3}$: $na - a = -3$, so $a = -\frac{1}{3}, n = 10$

Proofs using Binomial Theorem

ⓘ Binomial Coefficient Notation

To make expressions look simpler, we can use **binomial coefficient notation**:

$$^nC_r = \binom{n}{r}$$

Therefore, the binomial theorem can be written as:

$$(x+y)^n = \binom{n}{0}x^n + \binom{n}{1}x^{n-1}y + \binom{n}{2}x^{n-2}y^2 + \cdots + \binom{n}{n-1}xy^{n-1} + \binom{n}{n}y^n$$

ⓘ Combinations Proofs

You've already seen how to prove many combinations identities algebraically using the combinations formula or a combinatorial argument.

You'll now learn two more methods of proof:

- Substituting into the Binomial Theorem
- Equating coefficients

ⓟ **Prove** identities by substituting into the Binomial Theorem.

You have proved the identity below by counting the total number of subsets possible from a set of size n:

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots \binom{n}{n-1} + \binom{n}{n} = 2^n$$

We can prove this using the binomial theorem as well:

Starting with

$$(x+y)^n = \binom{n}{0}x^n + \binom{n}{1}x^{n-1}y + \binom{n}{2}x^{n-2}y^2 + \cdots + \binom{n}{n-1}xy^{n-1} + \binom{n}{n}y^n$$

Let $x = 1, y = 1$:

Prove identities by equating coefficients.

Using

$$\left(x+\frac{1}{x}\right)^n\left(x+\frac{1}{x}\right)^n=\left(x+\frac{1}{x}\right)^{2n}$$

Prove that $\binom{n}{0}^2+\binom{n}{1}^2+\binom{n}{2}^2\ldots+\binom{n}{n}^2=\binom{2n}{n}$ by equating the constant terms.

Find constant term of RHS:

The rth term of $\left(x+\frac{1}{x}\right)^{2n}$ is:

Let the power of x be 0 and solve for r.

Therefore, the constant term is:

Find constant term of LHS:

Find the rth term of $\left(x+\frac{1}{x}\right)^n$ and simplify to show that the expansion is:

$$\binom{n}{0}x^n+\binom{n}{1}x^{n-2}+\binom{n}{2}x^{n-4}+\cdots\binom{n}{n-1}x^{2-n}+\binom{n}{n}x^{-n}$$

(proof continues on next page)

For $\left(x+\frac{1}{x}\right)^n\left(x+\frac{1}{x}\right)^n$, the second factor expands identically:

$$\left(x+\frac{1}{x}\right)^n\left(x+\frac{1}{x}\right)^n = \left[\binom{n}{0}x^n+\binom{n}{1}x^{n-2}+\cdots\binom{n}{n-1}x^{2-n}+\binom{n}{n}x^{-n}\right]\left[\binom{n}{0}x^n+\binom{n}{1}x^{n-2}+\cdots\binom{n}{n-1}x^{2-n}+\binom{n}{n}x^{-n}\right]$$

When multiplying these, constant terms arise only when powers ……, for example:

$\binom{n}{0}x^n \times \binom{n}{n}x^{-n} = \binom{n}{0}\binom{n}{n}$

$\binom{n}{1}x^{n-2} \times \binom{n}{n-1}x^{2-n} = (\ )(\ \ \ \ )$

$\binom{n}{2}x^{(\ \ \ )} \times (\ \ \ \ )x^{(\ \ \ )} = (\ )(\ \ \ \ )$

The constant term comes from adding up all these pairs, so the constant term is:

Now consider the symmetric property: $\binom{n}{r} = \binom{n}{n-r}$

We can rewrite the constant term as:

We have therefore proven the identity.

$$\binom{n}{0}^2+\binom{n}{1}^2+\binom{n}{2}^2\ldots+\binom{n}{n}^2 = \binom{2n}{n}$$

This identity means that if the entries of any row are squared and added, the sum is the middle entry in the row twice as far down.

Row 0	1
Row 1	1 1
Row 2	1 2 1
Row 3	$1^2 + 3^2 + 3^2 + 1^2$
Row 4	1 4 6 4 1
Row 5	1 5 10 10 5 1
Row 6	1 6 15 **20** 15 6 1

Prove identities by combinatorial argument.

We will prove the same identity using a combinatorial argument.

$$\binom{n}{0}^2 + \binom{n}{1}^2 + \binom{n}{2}^2 \dots + \binom{n}{n}^2 = \binom{2n}{n}$$

The RHS represents choosing …… objects from ……, which we write as ……

We can make the same choice by dividing $2n$ objects into two distinct groups, A and B, then choosing r objects from one group, and $n - r$ objects from the other.

- Take 0 from A, n from B: $\binom{n}{0}\binom{n}{n} = \binom{n}{0}^2$
- Take 1 from A, $n - 1$ from B: $\binom{n}{1}\binom{n}{n-1} = \binom{n}{1}^2$
- Take 2 from A, …… from B:

…

All the possibilities add up to:

Since both sides count the same thing, we have proven that:

$$\binom{n}{0}^2 + \binom{n}{1}^2 + \binom{n}{2}^2 \dots + \binom{n}{n}^2 = \binom{2n}{n}$$

1 Consider the expansion for

$$(1 + x)^4 = {}^4C_0 + {}^4C_1x + {}^4C_2x^2 + {}^4C_3x^3 + {}^4C_4x^4 \quad (*)$$

Use it to prove the following results and interpret the result in terms of the row 1 4 6 4 1 on Pascal's triangle.

a By substituting $x = 1$, show that ${}^4C_0 + {}^4C_1 + {}^4C_2 + {}^4C_3 + {}^4C_4 = 2^4$.

$(1 + 1)^4 = {}^4C_0 + {}^4C_1 + {}^4C_2 + {}^4C_3 + {}^4C_4$

$2^4 = {}^4C_0 + {}^4C_1 + {}^4C_2 + {}^4C_3 + {}^4C_4$

This is the sum of all entries in row 1 4 6 4 1.

b By substituting $x = -1$, show that ${}^4C_0 + {}^4C_2 + {}^4C_4 = {}^4C_1 + {}^4C_3$.

$(1 - 1)^4 = {}^4C_0 - {}^4C_1 + {}^4C_2 - {}^4C_3 + {}^4C_4$

$0 = {}^4C_0 - {}^4C_1 + {}^4C_2 - {}^4C_3 + {}^4C_4$

Therefore, ${}^4C_0 + {}^4C_2 + {}^4C_4 = {}^4C_1 + {}^4C_3$

Alternating sum equals zero, so sum of even-positioned entries equals sum of odd-positioned entries.

c Hence, by using the result of part **a** and **b**, show that ${}^4C_0 + {}^4C_2 + {}^4C_4 = 2^3$.

From **a:** ${}^4C_0 + {}^4C_1 + {}^4C_2 + {}^4C_3 + {}^4C_4 = 2^4$

From **b:** ${}^4C_0 + {}^4C_2 + {}^4C_4 = {}^4C_1 + {}^4C_3$

Substituting **b** into **a**:

Therefore $2({}^4C_0 + {}^4C_2 + {}^4C_4) = 2^4$

So ${}^4C_0 + {}^4C_2 + {}^4C_4 = 2^3$

The sum of the even positioned terms of the row is half the sum of the whole row.

2 $(1 + x)^n = {}^nC_0 + {}^nC_1x + \ldots + {}^nC_nx^n$ $(*)$ for $n \geq 1$

Use it to prove the following results and interpret the result in terms of a general row n in Pascal's triangle.

a By substituting $x = 1$, show that ${}^nC_0 + {}^nC_1 + \ldots + {}^nC_n = 2^n$.

$(1 + 1)^n = {}^nC_0 + {}^nC_1 + \ldots + {}^nC_n$

$2^n = {}^nC_0 + {}^nC_1 + \ldots + {}^nC_n$

Sum of row n in Pascal's triangle is 2^n

b By substituting $x = -1$, show that ${}^nC_0 + {}^nC_2 + {}^nC_4 + \ldots = {}^nC_1 + {}^nC_3 + {}^nC_5 + \ldots$

$(1 - 1)^n = {}^nC_0 - {}^nC_1 + {}^nC_2 - {}^nC_3 + \ldots$

$0 = {}^nC_0 - {}^nC_1 + {}^nC_2 - {}^nC_3 + \ldots$

Therefore, ${}^nC_0 + {}^nC_2 + {}^nC_4 + \ldots = {}^nC_1 + {}^nC_3 + {}^nC_5 + \ldots$

The sum of the even position terms is equal to the sum of the odd position terms

c Hence, by using the result of parts **a** and **b**, show that ${}^nC_0 + {}^nC_2 + {}^nC_4 + \ldots = 2^{n-1}$.

${}^nC_0 + {}^nC_1 + \ldots + {}^nC_n = 2^n$

${}^nC_0 + {}^nC_2 + {}^nC_4 + \ldots = {}^nC_1 + {}^nC_3 + {}^nC_5 + \ldots$

Therefore $2({}^nC_0 + {}^nC_2 + {}^nC_4 + \ldots) = 2^n$

So ${}^nC_0 + {}^nC_2 + {}^nC_4 + \ldots = 2^{n-1}$

The sum of the even positioned terms of the row is half the sum of t

3 Consider the expansion: $(1 + x)^{2n} = {}^{2n}C_0 + {}^{2n}C_1x + \ldots + {}^{2n}C_{2n}x^{2n}$ $(*)$

a Show that ${}^{2n}C_0 + {}^{2n}C_1 + \ldots + {}^{2n}C_{2n} = 2^{2n}$.

Substitute x = 1 into (*):

$(1 + 1)^{2n} = {}^{2n}C_0 + {}^{2n}C_1 + \ldots + {}^{2n}C_{2n}$

$2^{2n} = {}^{2n}C_0 + {}^{2n}C_1 + \ldots + {}^{2n}C_{2n}$

b Show that ${}^{2n}C_1 + {}^{2n}C_3 + {}^{2n}C_5 + \ldots + {}^{2n}C_{2n-1} = 2^{2n-1}$.

Substitute $x = -1$ *into* $(*)$:

$0 = {}^{2n}C_0 - {}^{2n}C_1 + {}^{2n}C_2 - \ldots + {}^{2n}C_{2n}$

Therefore, ${}^{2n}C_1 + {}^{2n}C_3 + {}^{2n}C_5 + \ldots + {}^{2n}C_{2n-1} = {}^{2n}C_0 + {}^{2n}C_2 + \ldots + {}^{2n}C_{2n}$

From **a**, their sum is 2^{2n}, so each equals side is half, 2^{2n-1}.

4 $S = \{A, B, C, D, E\}$

a List all the subsets of $\{A, B, C, D, E\}$ containing 2 letters.

{A, B}, {A, C}, {A, D}, {A, E}, {B, C}, {B, D}, {B, E}, {C, D}, {C, E}, {D, E}

10 subsets in total.

b List all the subsets of $\{A, B, C, D, E\}$ containing 3 letters.

{A, B, C}, {A, B, D}, {A, B, E}, {A, C, D}, {A, C, E}, {A, D, E}, {B, C, D}, {B, C, E}, {B, D, E}, {C, D, E}

10 subsets in total.

c Consider the subset $\{A, B\}$. What letters have been omitted from the subset?

C, D, E have been omitted.

d Use the definition of ${}^{n}C_{r}$ to explain why ${}^{5}C_{2} = {}^{5}C_{3}$.

${}^{5}C_{2}$ = number of ways to choose 2 letters from 5

${}^{5}C_{3}$ = number of ways to choose 3 letters from 5

Choosing 2 letters to include is equivalent to choosing 3 letters to exclude.

Each 2-letter subset corresponds to exactly one 3-letter subset (its complement), e.g. {A B C} $\equiv \{D, E\}$

Therefore, ${}^{5}C_{2} = {}^{5}C_{3}$.

e Explain why ${}^{n}C_{r} = {}^{n}C_{n-r}$ for any whole numbers n and r with $r \leq n$.

from a total of n elements, choosing r items to include is equivalent to choosing $(n - r)$ items to exclude.

Each r-member subset corresponds to exactly one $(n - r)$-member subset (its complement).

Therefore ${}^{n}C_{r} = {}^{n}C_{n-r}$.

5 $S = \{A, B, C, D\}$

a Write down all the subsets of S, then explain why there are exactly 2^4 of them.

0-letter subsets: {}

1-letter subsets: {A}, {B}, {C}, {D}

2-letter subsets: {A, B}, {A, C}, {A, D}, {B, C}, {B, D}, {C, D}

3-letter subsets: {A, B, C}, {A, B, D}, {A, C, D}, {B, C, D}

4-letter subsets: {A, B, C, D}

Total: 1 + 4 + 6 + 4 + 1 = 16 = 2^4

For each element, there are 2 choices: include it or exclude it.

Since there are 4 elements and choices are independent: $2 \times 2 \times 2 \times 2 = 2^4$ subsets.

b Hence explain why $\binom{4}{0} + \binom{4}{1} + \binom{4}{2} + \binom{4}{3} + \binom{4}{4} = 2^4$.

$\binom{4}{0}$ counts 0-letter subsets (1 subset)

$\binom{4}{1}$ counts 1-letter subsets (4 subsets)

$\binom{4}{2}$ counts 2-letter subsets (6 subsets)

$\binom{4}{3}$ counts 3-letter subsets (4 subsets)

$\binom{4}{4}$ counts 4-letter subsets (1 subset)

These cases make up all possible subsets, so their sum equals the total number of subsets = 2^4.

c Explain why $\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \binom{n}{3} + \binom{n}{4}) = 2^n$, for any whole number n.

For an n-element set, each element has 2 choices (include or exclude), giving 2^n total subsets.

$\binom{n}{0}$ counts subsets with 0 elements

$\binom{n}{1}$ counts subsets with 1 element

...

$\binom{n}{n}$ counts subsets with n elements

These cases make up all possible subsets, so $\binom{n}{0} + \binom{n}{1} + \ldots + \binom{n}{n} = 2^n$.

6 $S = \{A, B, C, D, E\}$ and $U = \{A, B, C, D\}$

a Write down all the 3-letter subsets of S that do not contain E.

{A, B, C}, {A, B, D}, {A, C, D}, {B, C, D}

b Explain why this is a list of all the 3-letter subsets of U.

U is a subset of S that doesn't contain E.

Since E is not in U, all 3-letter subsets of U automatically do not contain E.

c Write down all the 3-letter subsets of S that contain E.

{A, B, E}, {A, C, E}, {A, D, E}, {B, C, E}, {B, D, E}, {C, D, E}

d Explain how to pair them up with all the 2-letter subsets of U.

Each 3-letter subset containing E has the form {X, Y, E} where {X, Y} is a 2-letter subset of U.

Remove E from each 3-letter subset to get the corresponding 2-letter subset of U:

{A, B, E} ↔ {A, B}

{A, C, E} ↔ {A, C}

{A, D, E} ↔ {A, D}

{B, C, E} ↔ {B, C}

{B, D, E} ↔ {B, D}

{C, D, E} ↔ {C, D}

e Hence explain why ${}^4C_2 + {}^4C_3 = {}^5C_3$.

${}^5C_3$ = number of 3-letter subsets of S

These split into two categories:

Those not containing E: these are exactly the 3-letter subsets of U, counted by ${}^4C_3$

Those containing E: these correspond to 2-letter subsets of U, counted by ${}^4C_2$

Therefore, ${}^4C_2 + {}^4C_3 = {}^5C_3$.

f Explain why ${}^nC_{r-1} + {}^nC_r = {}^{n+1}C_r$ for any whole numbers n and r with $1 \leq r \leq n$.

${}^{n+1}C_r$ = number of r-member subsets of an $(n + 1)$-element set

Label one element as "X"

This splits the cases into two disjoint categories:

r-member subsets not containing X: these are r-member subsets of the remaining n elements, counted by ${}^nC_r$

r-member subsets containing X: remove X to get $(r - 1)$-member subsets of the remaining n elements, counted by ${}^nC_{r-1}$

Therefore ${}^nC_{r-1} + {}^nC_r = {}^{n+1}C_r$.

Exam Questions

1 2021 HSC Extension 1 Band 3

Expand and simplify $(2a - b)^4$

$= 16a^4 + 4(8a^3)(-b) + 6(4a^2)(b^2) + 4(2a)(-b^3) + b^4$

$= 16a^4 - 32a^3b + 24a^2b^2 - 8ab^3 + b^4$

2 2017 HSC Extension 1 Band 4

When expanded, which expression has a non-zero constant term?

A. $\left(x + \frac{1}{x^2}\right)^7$

B. $\left(x^2 + \frac{1}{x^3}\right)^7$

C. $\left(x^3 + \frac{1}{x^4}\right)^7$

D. $\left(x^4 + \frac{1}{x^5}\right)^7$

Option A:

${}^7C_k \cdot x^{7-k} \cdot x^{-2k} = {}^7C_k \cdot x^{7-3k}$

Constant when 7-3k = 0

k = 7/3 (not integer, no constant term)

Option B:

${}^7C_k \cdot x^{2(7-k)} \cdot x^{-3k} = {}^7C_k \cdot x^{14-5k}$

Constant when 14-5k = 0

k = 14/5 (not integer, no constant term)

Option C:

${}^7C_k \cdot x^{3(7-k)} \cdot x^{-4k} = {}^7C_k \cdot x^{21-7k}$

Constant when $21 - 7k = 0, k = 3$

Constant term $= {}^7C_3 = 35$

Answer: C

3 2016 HSC Extension 2 Band 4

Let $p(x) = 1 + x + x^2 + x^3 + \ldots + x^{12}$.

What is the coefficient of x^8 in the expansion of $p(x + 1)$?

$p(x + 1) = 1 + (x + 1) + (x + 1)^2 + \ldots + (x + 1)^{12}$

Coefficient of x^8 comes from terms where $(x + 1)^n$ contributes x^8

From $(x + 1)^n$, coefficient of x^8 is ${}^nC_8$

Need $n \geq 8$, so $n = 8, 9, 10, 11, 12$

Coefficient of $x^8 = {}^8C_8 + {}^9C_8 + {}^{10}C_8 + {}^{11}C_8 + {}^{12}C_8$

$= 1 + 9 + 45 + 165 + 495$

$= 715$

4 2019 HSC Extension 1 Band 4

In the expansion of $(5x + 2)^{20}$, the coefficients of x^{r} and x^{r+1} are equal.
What is the value of r?

r^{th} term: ${}^{20}C_r \cdot 2^{20-r} \cdot (5x)^r = {}^{20}C_r \cdot 2^{20-r} \cdot 5^r \cdot x^r$

$r+1^{\text{th}}$ term ${}^{20}C_{r+1} \cdot 2^{19-r} \cdot 5^{r+1} \cdot x^{r+1}$

Equating coefficients:

$${}^{20}C_r \cdot 2^{20-r} \cdot 5^r = {}^{20}C_{r+1} \cdot 2^{19-r} \cdot 5^{r+1}$$

$$\frac{20!}{r!\,(20-r)!} \cdot 2^{20-r} \cdot 5^r = \frac{20!}{(r+1)!\,(19-r)!} \cdot 2^{19-r} \cdot 5^{r+1}$$

$$\frac{2}{20-r} = \frac{5}{r+1}$$

$$2(r+1) = 5(20-r)$$

$$2r + 2 = 100 - 5r$$

$$7r = 98$$

$$r = 14$$

5 2022 HSC Extension 1 Band 4

Find the coefficients of x^2 and x^3 in the expansion of $\left(1 - \frac{x}{2}\right)^8$

$${}^{8}C_k \cdot (1)^{8-k} \cdot \left(-\frac{x}{2}\right)^k = {}^{8}C_k \cdot \left(-\frac{1}{2}\right)^k \cdot x^k$$

Coefficient of x^2 $(k = 2)$:

$${}^{8}C_2 \cdot \left(-\frac{1}{2}\right)^2 = 7$$

Coefficient of x³ (k = 3):

$${}^{8}C_3 \cdot \left(-\frac{1}{2}\right)^3 = -7$$

6 2023 HSC Extension 1 Band 4

It is known that ${}^{\text{n}}C_{\text{r}} = {}^{\text{n}-1}C_{\text{r}-1} + {}^{\text{n}-1}C_{\text{r}}$ for all integers such that $1 \leq r \leq n - 1$.

(Do NOT prove this.)
Find ONE possible set of values for p and q such that ${}^{2022}C_{80} + {}^{2022}C_{81} + {}^{2023}C_{1943} = {}^{p}C_{q}$

Using ${}^{\text{n}}C_{\text{r}} = {}^{\text{n}-1}C_{\text{r}-1} + {}^{\text{n}-1}C_{\text{r}}$:

$${}^{2022}C_{80} + {}^{2022}C_{81} = {}^{2023}C_{81} \ldots (1)$$

Using ${}^{\text{n}}C_{\text{r}} = {}^{\text{n}}C_{\text{n}-\text{r}}$:

$${}^{2023}C_{1943} = {}^{2023}C_{80} \ldots (2)$$

From (1) and (2):

$${}^{2022}C_{80} + {}^{2022}C_{81} + {}^{2023}C_{1943} = {}^{2023}C_{81} + {}^{2023}C_{80}$$

Using ${}^{\text{n}-1}C_{\text{r}-1} + {}^{\text{n}-1}C_{\text{r}} = {}^{\text{n}}C_{\text{r}}$:

$${}^{2023}C_{80} + {}^{2023}C_{81} = {}^{2024}C_{81}$$

$$p = 2024, q = 81$$

7 HSC Extension 1 Sample Question Band 4

Find the coefficient of x^4 in the expansion of $\left(x^2 - \frac{3}{x}\right)^5$

$${}^5C_k \cdot (x^2)^{5-k} \cdot \left(-\frac{3}{x}\right)^k$$

$$= {}^5C_k \cdot x^{10-2k} \cdot (-3)^k \cdot x^{-k}$$

$$= {}^5C_k \cdot x^{10-3k} \cdot (-3)^k$$

x⁴ occurs when:

$$10 - 3k = 4$$

$$k = 2$$

$${}^5C_2 \cdot (-3)^2 = 90$$

8 HSC Extension 1 Sample Question Band 4

By using the fact that $(1 + x)^{11} = (1 + x)^3(1 + x)^8$, show that:

$$\binom{11}{5} = \binom{8}{5} + \binom{3}{1}\binom{8}{4} + \binom{3}{2}\binom{8}{3} + \binom{8}{2}$$

General term of $(1 + x)^{11}$:

$${}^{11}C_k \cdot x^k$$

$\therefore$ ${}^{11}C_5$ is the coefficient of x^5

$$(1 + x)^3 = {}^3C_0 + {}^3C_1x + {}^3C_2x^2 + {}^3C_3x^3$$

$$(1 + x)^8 = {}^8C_0 + {}^8C_1x + {}^8C_2x^2 + {}^8C_3x^3 + {}^8C_4x^4 + {}^8C_5x^5 + \ldots$$

Coefficient of x^5 in $(1 + x)^3(1 + x)^8$ occurs when sum of indices = 5

$$= {}^3C_0 \cdot {}^8C_5 + {}^3C_1 \cdot {}^8C_4 + {}^3C_2 \cdot {}^8C_3 + {}^3C_3 \cdot {}^8C_2$$

$$= {}^8C_5 + {}^3C_1 \cdot {}^8C_4 + {}^3C_2 \cdot {}^8C_3 + {}^8C_2$$

Equating coefficients:

$${}^{11}C_5 = {}^8C_5 + {}^3C_1 \cdot {}^8C_4 + {}^3C_2 \cdot {}^8C_3 + {}^8C$$

9 HSC Extension 1 Sample Question Band 5

Using $(1 + x)^4(1 + x)^9 = (1 + x)^{13}$, show that

$$^9C_4 + {}^4C_1 \cdot {}^9C_3 + {}^4C_2 \cdot {}^9C_2 + {}^4C_3 \cdot {}^9C_1 + 1 = {}^{13}C_4$$

General term of $(1 + x)^{13}$:

$^{13}C_k \cdot x^k$

$\therefore$ $^{13}C_4$ is coefficient of x^4

$(1 + x)^4 = {}^4C_0 + {}^4C_1x + {}^4C_2x^2 + {}^4C_3x^3 + {}^4C_4x^4$

$(1 + x)^9 = {}^9C_0 + {}^9C_1x + {}^9C_2x^2 + {}^9C_3x^3 + {}^9C_4x^4 + \ldots$

Coefficient of x^4 in $(1 + x)^4(1 + x)^9$ when sum of indices = 4

$= {}^4C_0 \cdot {}^9C_4 + {}^4C_1 \cdot {}^9C_3 + {}^4C_2 \cdot {}^9C_2 + {}^4C_3 \cdot {}^9C_1 + {}^4C_4 \cdot {}^9C_0$

$= {}^9C_4 + {}^4C_1 \cdot {}^9C_3 + {}^4C_2 \cdot {}^9C_2 + {}^4C_3 \cdot {}^9C_1 + 1$

Equating coefficients:

$^9C_4 + {}^4C_1 \cdot {}^9C_3 + {}^4C_2 \cdot {}^9C_2 + {}^4C_3 \cdot {}^9C_1 + 1 = {}^{13}C_4$

10 2012 HSC Extension 1 Band 6

a Use the binomial theorem to find an expression for the constant term in the expansion of

$$\left(2x^3 - \frac{1}{x}\right)^{12}$$

General term:

$^2C_k \cdot (2x^3)^{12-k} \cdot \left(-\frac{1}{x}\right)^k$

$= {}^{12}C_k \cdot (-1)^k \cdot 2^{12-k} \cdot x^{36-3k} \cdot x^{-k}$

$= {}^{12}C_k \cdot (-1)^k \cdot 2^{12-k} \cdot x^{36-4k}$

Constant term when $36 - 4k = 0$

$k = 9$

Constant term: $^{12}C_9 \cdot (-1)^9 \cdot 2^3$

$= -1760$

b For what values of n does $\left(2x^3 - \frac{1}{x}\right)^n$ have a non-zero constant term?

General term: $^nC_k \cdot (-1)^k \cdot 2^{n-k} \cdot x^{3n-4k}$

Constant term when $3n - 4k = 0$

$k = \frac{3n}{4}$

Since n and k must be integers, n must be a multiple of 4

11 2020 HSC Extension 1 Band 6

a Use the identity $(1 + x)^{2n} = (1 + x)^n(1 + x)^n$ to show that

$$^{2n}C_n = (^nC_0)^2 + (^nC_1)^2 + \ldots + (^nC_n)^2$$

where n is a positive integer.

Coefficient of x^n in $(1 + x)^{2n} = {}^{2n}C_n$

Coefficient of x^n in $(1 + x)^n(1 + x)^n$:

$= {}^nC_0 \cdot {}^nC_n + {}^nC_1 \cdot {}^nC_{n-1} + \ldots + {}^nC_n \cdot {}^nC_0$

$= (^nC_0)^2 + (^nC_1)^2 + \ldots + (^nC_n)^2 \; (using \; {}^nC_r = {}^nC_{n-r})$

Equating coefficients:

$^{2n}C_n = (^nC_0)^2 + (^nC_1)^2 + \ldots + (^nC_n)^2$

b A club has $2n$ members, with n women and n men. A group consisting of an even number $(0, 2, 4, \ldots, 2n)$ of members is chosen, with the number of men equal to the number of women.

Show, giving reasons, that the number of ways to do this is $^{2n}C_n$.

Number of men equal to number of women, so the club can have group sizes of each gender 0 up to n

Ways to choose group of 0 men/women: ${}^nC_0 \times {}^nC_0 = (^nC_0)^2$

Ways to choose group of 1 man/woman: ${}^nC_1 \times {}^nC_1 = (^nC_1)^2$

…

Ways to choose group of n men/women: ${}^nC_n \times {}^nC_n = (^nC_n)^2$

Total = $(^nC_0)^2 + (^nC_1)^2 + \ldots + (^nC_n)^2 = {}^{2n}C_n$

c From the group chosen in part **b**, one of the men and one of the women are selected as leaders.

Show, giving reasons, that the number of ways to choose the even number of people and then the leaders is

$$1^2(^nC_1)^2 + 2^2(^nC_2)^2 + \ldots + n^2(^nC_n)^2$$

For each group of k men/women, choose a leader in ${}^kC_1 = k$ ways

0: can't choose a leader.

1: ${}^nC_1 \times 1 \times {}^nC_1 \times 1 = 1^2(^nC_1)^2$

2: ${}^nC_2 \times 2 \times {}^nC_2 \times 2 = 2^2(^nC_2)^2$

…

n: ${}^nC_n \times n \times {}^nC_n \times n = n^2(^nC_n)^2$

Therefore, sum of all ways in all possible group sizes is

$1^2(^nC_1)^2 + 2^2(^nC_2)^2 + \ldots + n^2(^nC_n)^2$

d The process is now reversed so that the leaders, one man and one woman, are chosen first. The rest of the group is then selected, still made up of an equal number of women and men.
By considering this reversed process and using part **b** find a simple expression for the sum in part **c**.

For each group of k men/women, choose a leader of each gender first ${}^{n}C_1 = n$, then choose other members of each gender ${}^{n-1}C_{k-1}$ ways.

$$1{:}\ n \times n \times {}^{n-1}C_0 \times {}^{n-1}C_0 = n^2({}^{n-1}C_0)^2$$
$$2{:}\ n \times n \times {}^{n-1}C_1 \times {}^{n-1}C_1 = n^2({}^{n-1}C_1)^2$$
$$\dots$$
$$n{:}\ n \times n \times {}^{n-1}C_{n-1} \times {}^{n-1}C_{n-1} = n^2({}^{n-1}C_{n-1})^2$$

Therefore, sum of all ways in all possible group sizes is

$$n^2({}^{n-1}C_0)^2 + n^2({}^{n-1}C_1)^2 + \cdots + n^2({}^{n-1}C_{n-1})^2$$
$$= n^2[({}^{n-1}C_0)^2 + ({}^{n-1}C_1)^2 + \cdots + ({}^{n-1}C_{n-1})^2]$$

Using $({}^{n}C_0)^2 + ({}^{n}C_1)^2 + \dots + ({}^{n}C_n)^2 = {}^{2n}C_n$

$$n^2[({}^{n-1}C_0)^2 + ({}^{n-1}C_1)^2 + \cdots + ({}^{n-1}C_{n-1})^2] = n^2 \times {}^{2(n-1)}C_{n-1}$$
$$= n^2 \times {}^{2n-2}C_{n-1}$$

Since this counts the same thing as part **c,**

$$1^2({}^{n}C_1)^2 + 2^2({}^{n}C_2)^2 + \dots + n^2({}^{n}C_n)^2 = n^2 \times {}^{2n-2}C_{n-1}$$

Challenge Exercise

1

a In the expansion of $(2 + 3x)^n$, the coefficients of x^5 and x^6 are in the ratio 4 : 9. Find the value of n.

Coefficient of x^5: ${}^nC_5\ 2^{n-5}\ 3^5$

Coefficient of x^6: ${}^nC_6\ 2^{n-6}\ 3^6$

Ratio: $({}^nC_5\ 2^{n-5}\ 3^5)/({}^nC_6\ 2^{n-6}\ 3^6) = 4/9$

$({}^nC_5/{}^nC_6)(2/3) = 4/9$

$((n-5)/(6))(2/3) = 4/9$

$(n-5)/9 = 2/3$, so $n = 14$

b In the expansion of $\left(x + \frac{1}{x}\right)^{40}\left(x - \frac{1}{x}\right)^{40}$, find the term independent of x. Give your answer in the form ${}^nC_r$, and also correct to four significant figures.

The expansion is equivalent to $\left(x^2 - \frac{1}{x^2}\right)^{40}$

General term: ${}^{40}C_r(x^2)^r(x^{-2})^{40-r}$

$= {}^{40}C_r x^{4r-80}$

For $x^0, 4r - 80 = 0$ so $r = 20$

${}^{40}C_{20} = 1.378 \times 10^{11}$

2

a Use the binomial theorem to show that $7^n + 2$ is divisible by 3, where n is a positive integer. (Hint: Write 7 = 6 + 1.)

$7^n = (6 + 1)^n = 6^n + {}^nC_1\,6^{n-1} + \ldots + {}^nC_{n-1}\,6 + 1$

$7^n + 2 = 6^n + {}^nC_1\,6^{n-1} + \ldots + {}^nC_{n-1}\,6 + 3$

$= 3(2 \cdot 6^{n-1} + n \cdot 6^{n-2} + \ldots + 2n + 1)$

divisible by 3

b Use the binomial theorem to show that $5^n + 3$ is divisible by 4, where n is a positive integer.

$5^n = (4 + 1)^n = 4^n + {}^nC_1\,4^{n-1} + \ldots + {}^nC_{n-1}\,4 + 1$

$5^n + 3 = 4^n + {}^nC_1\,4^{n-1} + \ldots + 4n + 4$

$= 4(4^{n-1} + n \cdot 4^{n-2} + \ldots + n + 1),$

divisible by 4

3

a Expand and simplify $(x + y)^6 + (x - y)^6$.

$(x + y)^6 = x^6 + 6x^5y + 15x^4y^2 + 20x^3y^3 + 15x^2y^4 + 6xy^5 + y^6$

$(x - y)^6 = x^6 - 6x^5y + 15x^4y^2 - 20x^3y^3 + 15x^2y^4 - 6xy^5 + y^6$

$(x + y)^6 + (x - y)^6 = 2x^6 + 30x^4y^2 + 30x^2y^4 + 2y^6$

b Hence (and without a calculator) prove that

$$5^6 + 5^5 \times 3^3 + 5^3 \times 3^5 + 3^6 = 2^5(2^{12} + 1)$$

Let $x = 5, y = 3$

$(5 + 3)^6 + (5 - 3)^6 = 2(5^6) + 30(5^4)(3^2) + 30(5^2)(3^4) + 2(3^6)$

$8^6 + 2^6 = 2(5^6) + 30(5^4)(9) + 30(25)(81) + 2(3^6)$

$2^6(2^{12} + 1) = 2(5^6 + 5^5(3^3) + 5^3(3^5) + 3^6)$

$\therefore 5^6 + 5^5(3^3) + 5^3(3^5) + 3^6 = 2^5(2^{12} + 1)$

4 Find the coefficients of x and x^{-3} in the expansion of $\left(3x - \frac{a}{x}\right)^5$.
Hence find the values of a if these coefficients are in the ratio 2 : 1.

r^{th} term: ${}^5C_r\,(3x)^{5-r}\left(-\frac{a}{x}\right)^r = {}^5C_r\,3^{5-r}\,(-a)^r\,x^{5-2r}$

For x, let $5 - 2r = 1,\ r = 2$: ${}^5C_2\,3^3\,(-a)^2 = 270a^2$

For x^{-3}, let $5 - 2r = -3,\ r = 4$: ${}^5C_4\,3\,(-a)^4 = 15a^4$

$\frac{270a^2}{15a^4} = \frac{2}{1}$

$a^2 = 9, a = \pm 3$

5 The coefficients of the terms in a^3 and a^{-3} in the expansion of $\left(ma + \frac{n}{a^2}\right)^6$ are equal, where m and n are non-zero real numbers.
Prove that $m^2 : n^2 = 10 : 3$.

r^{th} term: ${}^6C_r\,(ma)^{6-r}\left(\frac{n}{a^2}\right)^r = {}^6C_r\,m^{6-r}\,n^r\,a^{6-3r}$

For a^3, let $6 - 3r = 3$, so $r = 1$: ${}^6C_1\,m^5n$

For a^{-3}, let $6 - 3r = -3$, so $r = 3$: ${}^6C_3\,m^3n^3$

Equal: ${}^6C_1\,m^5n = {}^6C_3\,m^3n^3$

$6m^5n = 20m^3n^3$

$3m^2 = 10n^2$

$\frac{m^2}{n^2} = \frac{10}{3}$

6 The Hockey Stick Identity

Start at the top 1 of any column of Pascal's triangle, and go vertically down any number of rows.

Then the sum of these numbers is the number in the next row down and to the right.

Consider row 2 of the triangle:

Row 0	1						
Row 1	1	1					
Row 2	1	2	**1**				
Row 3	1	3	**3**	1			
Row 4	1	4	**6**	4	1		
Row 5	1	5	**10**	10	5	1	
Row 6	1	6	**15**	20	15	6	1
Row 7	1	7	21	**35**	35	21	7	1

$${}^2C_2 + {}^3C_2 + {}^4C_2 + {}^5C_2 + {}^6C_2 = 1 + 3 + 6 + 10 + 15 = 35 = {}^7C_3 \ (*)$$

The general form is:

$${}^rC_r + {}^{r+1}C_r + {}^{r+2}C_r + \ldots + {}^nC_r = {}^{n+1}C_{r+1} \ (**)$$

a To prove the particular result (*), start with the right-hand side and use Pascal's Identity:

$${}^7C_3 = {}^6C_3 + {}^6C_2$$

then keep expanding the term with $r = 3$.

To complete the proof, you will need to use the fact that ${}^3C_3 = {}^2C_2$

$$\begin{aligned} {}^7C_3 &= {}^6C_3 + {}^6C_2 \\ &= {}^5C_3 + {}^5C_2 + {}^6C_2 \\ &= {}^4C_3 + {}^4C_2 + {}^5C_2 + {}^6C_2 \\ &= {}^3C_3 + {}^3C_2 + {}^4C_2 + {}^5C_2 + {}^6C_2 \\ &= {}^2C_2 + {}^3C_2 + {}^4C_2 + {}^5C_2 + {}^6C_2 \end{aligned}$$

b Hence generalise this to the proof of the identity $(**)$

$$\begin{aligned} {}^{n+1}C_{r+1} &= {}^nC_{r+1} + {}^nC_r \\ &= {}^{n-1}C_{r+1} + {}^{n-1}C_r + {}^nC_r \\ &= {}^{n-2}C_{r+1} + {}^{n-2}C_r + {}^{n-1}C_r + {}^nC_r \\ &= \ldots \\ &= {}^{r+1}C_{r+1} + {}^{r+1}C_r + {}^{r+2}C_r + \ldots + {}^nC_r \\ &= {}^rC_r + {}^{r+1}C_r + {}^{r+2}C_r + \ldots + {}^nC_r \ (\text{since } {}^{r+1}C_{r+1} = {}^rC_r) \end{aligned}$$

7 If the entries of any row are squared and added, the sum is the middle entry in the row twice as far down. We proved this in an investigation earlier, we will prove it again using a different expansion.

By comparing coefficients of x^n on both sides of the identity $(1 + x)^n(1 + x)^n = (1 + x)^{2n}$, show that

$$({}^nC_0)^2 + ({}^nC_1)^2 + ({}^nC_2)^2 + ({}^nC_3)^2 + \ldots + ({}^nC_n)^2 = {}^{2n}C_n$$

$LHS = (1 + x)^n(1 + x)^n$

$= ({}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \ldots + {}^nC_nx^n)({}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \ldots + {}^nC_nx^n)$

Coefficient of x^n on LHS:

${}^nC_0 \cdot {}^nC_n + {}^nC_1 \cdot {}^nC_{n-1} + {}^nC_2 \cdot {}^nC_{n-2} + \ldots + {}^nC_n \cdot {}^nC_0$

Using ${}^nC_r = {}^nC_{n-r}$:

$= ({}^nC_0)^2 + ({}^nC_1)^2 + ({}^nC_2)^2 + \ldots + ({}^nC_n)^2$

$RHS = (1 + x)^{2n}$

$= {}^{2n}C_0 + {}^{2n}C_1x + \ldots + {}^{2n}C_nx^n + \ldots + {}^{2n}C_{2n}x^{2n}$

Coefficient of x^n on RHS: ${}^{2n}C_n$

Therefore $({}^nC_0)^2 + ({}^nC_1)^2 + ({}^nC_2)^2 + \ldots + ({}^nC_n)^2 = {}^{2n}C_n$

8 In this question we consider binary words consisting only of the letters A and B.

a Consider a binary word consisting of $a + b$ letters, with A occuring a times and B occuring b times. Show that there are ${}^{a+b}C_a = {}^{a+b}C_b$ permutations of such a word.

Total positions: $a + b$

Choose a positions for letter A: ${}^{a+b}C_a$ ways

Remaining b positions automatically filled with B.

Therefore ${}^{a+b}C_a$ permutations.

Equivalently, choose b positions for letter B: ${}^{a+b}C_b$ ways.

Since both give the same count, ${}^{a+b}C_a = {}^{a+b}C_b$

b How many possible permutations are there of a binary word with $2n$ letters, if A and B both occur n times?

By substituting $a = n,\ b = n$:

${}^{2n}C_n$

c A word with $2n$ letters may be split down the middle into two words of n letters.
Consider the example where two A's fall in the first n-letter word.

i How many arrangements are there of the first n-letter binary word with two As?

Out of n positions, choose where to place 2 As.

Number of arrangements = ${}^nC_2$

ii How many arrangements are there of the second n-letter binary word with $n - 2$ A's and 2 B's?

Out of n positions, choose where to place 2 Bs.

Number of arrangements = ${}^nC_2$

iii Therefore, how many arrangements are there of a $2n$ letter binary word with two As in the first half and two Bs in the second half?

First half: ${}^nC_2$

Second half: ${}^nC_2$

Total arrangements = ${}^nC_2 \times {}^nC_2 = ({}^nC_2)^2$

d Hence give a combinatorial argument to prove

$$({}^nC_0)^2 + ({}^nC_1)^2 + ({}^nC_2)^2 + ({}^nC_3)^2 + \ldots + ({}^nC_n)^2 = {}^{2n}C_n$$

Consider a $2n$-letter binary word with n As and n Bs.

Total arrangements = ${}^{2n}C_n$ (from part b).

Split the word into two n-letter halves.

Suppose k A's fall in the first half (then $n - k$ A's in second half).

First half: k A's and $(n - k)$ Bs: ${}^nC_k$ arrangements

Second half: $(n - k)$ As and k Bs: ${}^nC_k$ arrangements

Total for this split: $({}^nC_k)^2$ arrangements

Sum over all possible values $k = 0, 1, 2, \ldots, n$:

${}^{2n}C_n = ({}^nC_0)^2 + ({}^nC_1)^2 + ({}^nC_2)^2 + \ldots + ({}^nC_n)^2$

9 Proofs involving Calculus

$$(1 + x)^4 = {}^4C_0 + {}^4C_1x + {}^4C_2x^2 + {}^4C_3x^3 + {}^4C_4x^4$$

a Differentiate both sides of the binomial expansion above.

$4(1 + x)^3 = {}^4C_1 + 2{}^4C_2x + 3{}^4C_3x^2 + 4{}^4C_4x^3$

b By substituting $x = 1$, show that ${}^4C_1 + 2({}^4C_2) + 3({}^4C_3) + 4({}^4C_4) = 4 \times 2^3$.

$4(2)^3 = {}^4C_1 + 2{}^4C_2 + 3{}^4C_3 + 4{}^4C_4$

$4 \times 2^3 = {}^4C_1 + 2{}^4C_2 + 3{}^4C_3 + 4{}^4C_4$

The sum of the products of a term and its position is the index of the row times half the sum of the whole row.

c By substituting $x = -1$, show that ${}^4C_1 - 2({}^4C_2) + 3({}^4C_3) - 4({}^4C_4) = 0$.

$4(0)^3 = {}^4C_1 - 2{}^4C_2 + 3{}^4C_3 - 4{}^4C_4$

$0 = {}^4C_1 - 2{}^4C_2 + 3{}^4C_3 - 4{}^4C_4$

The alternating sum of the products of a term and its position is zero

7 $(1 + x)^n = {}^nC_0 + {}^nC_1x + \ldots + {}^nC_nx^n$ for $n \geq 1$

a Differentiate both sides of the binomial expansion above.

$n(1 + x)^{n-1} = {}^nC_1 + 2{}^nC_2x + 3{}^nC_3x^2 + \ldots + n{}^nC_nx^{n-1}$

b By substituting $x = 1$, show that $1 \times {}^nC_1 + 2 \times {}^nC_2 + \ldots + n \times {}^nC_n = n2^{n-1}$.

$n(2)^{n-1} = {}^nC_1 + 2{}^nC_2 + 3{}^nC_3 + \ldots + n{}^nC_n$

The sum of the products of a term and its position is the index of the row times half the sum of the whole row.

c By substituting $x = -1$, show that $1 \times {}^nC_1 - 2 \times {}^nC_2 + \ldots + (-1)^{n-1}n \times {}^nC_n = 0$.

$n(0)^{n-1} = {}^nC_1 - 2{}^nC_2 + 3{}^nC_3 - \ldots + (-1)^{n-1}n{}^nC_n$

$0 = 1 \times {}^nC_1 - 2 \times {}^nC_2 + \ldots + (-1)^{n-1}n \times {}^nC_n$

The alternating sum of the products of a term and its position is zero

8 Consider the expansion: $(1 + x)^{2n} = {}^{2n}C_0 + {}^{2n}C_1x + \ldots + {}^{2n}C_{2n}x^{2n}$

a By differentiating both sides of the identity, show that:
$1 \times {}^{2n}C_1 + 2 \times {}^{2n}C_2 + \ldots + 2n \times {}^{2n}C_{2n} = n2^{2n}$.

$2n(1 + x)^{2n-1} = {}^{2n}C_1 + 2{}^{2n}C_2x + 3{}^{2n}C_3x^2 + \ldots + 2n{}^{2n}C_{2n}x^{2n-1}$

Substitute $x = 1$:

$2n(2)^{2n-1} = {}^{2n}C_1 + 2{}^{2n}C_2 + \ldots + 2n{}^{2n}C_{2n}$

$n2^{2n} = 1 \times {}^{2n}C_1 + 2 \times {}^{2n}C_2 + \ldots + 2n \times {}^{2n}C_{2n}$

b And show that $1 \times {}^{2n}C_1 - 2 \times {}^{2n}C_2 + \ldots + (-1)^{2n-1}2n \times {}^{2n}C_{2n} = 0$.

Substitute $x = -1$ into differentiated form:

$2n(0)^{2n-1} = {}^{2n}C_1 - 2{}^{2n}C_2 + 3{}^{2n}C_3 - \ldots + (-1)^{2n-1}2n{}^{2n}C_{2n}$

$0 = 1 \times {}^{2n}C_1 - 2 \times {}^{2n}C_2 + \ldots + (-1)^{2n-1}2n \times {}^{2n}C_{2n}$